napari-raman-widget

Detailed User and Architecture Guide

A control-by-control guide to the napari Raman sidebar, including preparation, napari layer behavior, selection-to-MDA dependencies, outputs, and safety checks.

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Architecture and Workflow Dependencies

Main components

The application combines napari visualization, Qt controls, Micro-Manager device control, Raman acquisition, coordinate transformation, autofocus, calibration, cell selection, stage positioning, and multidimensional acquisition. The sidebar organizes these capabilities into a sequence of expandable sections. Plot windows show spectra and calibration results, while the log window records acquisition progress.

Begin each hardware session with **Connect hardware**. The bottom status label reports the current action and confirms when each workflow is ready.

The selection-to-MDA contract

Run Raman MDA uses three prepared variables: `sources`, `autofocus_p`, and `new_seq`. Prepare these variables with one of the following selection workflows:

- **Run automated selection**;
- **Manual selection** (after the user clicks points in the created layers);
- **Center clicked cells** (optional conversion of a non-batch manual selection);
- **Generate grid** in the stage-grid section.

Before selection begins, live acquisition stops and the napari-micromanager MDA settings are prepared with axis order `t,p,c,z` and a centered Z plan. The recommended sequence is:

1. Connect hardware; this attempts to open the `napari-micromanager` dock.
2. In that MDA widget, define or verify stage positions and the basic sequence.
3. Use automated selection, manual selection, or stage-grid generation to create the source layers and revised sequence.
4. For manual selection, finish clicking the intended cells in the layers.
5. Configure **Run Raman MDA**, then start it.

The MDA dock provides the underlying position and sequence information used by the selection controls.

Coordinate conventions

Point and shape workflows use the *last* napari layer. Point workflows use the first point in that layer and its final two coordinates. For a point stored as (y, x), the transformer converts the normalized bright-field coordinate into galvanometer volts. Place the intended Points or Shapes layer last in the layer list.

Loading

This is the only section expanded at startup. Choose paths and the output folder before connecting, because connection loads the devices and changes the process working directory.

Control	Default / allowed	Meaning and use
Micro-Manager config (.cfg)	`test3.cfg`	Device configuration used for the device setup. The browse button filters for CFG files. During connection the widget also attempts a GFP-to-BF channel warm-up.
Transformer model (.json)	blank; example `model_2026-01-08.json`	Coordinate transformer used for bright-field pixel to Raman galvo conversion. Point collection, calibration, reference acquisition, mapping, and Raman MDA require it.
Vandermonde model (.json)	blank; example `vandermonde_model.json`	Pixel-to-stage model used only when cells are physically centered. Required by automated **Center cell** and **Center clicked cells**. It complements the Raman aiming transformer.
Output folder	blank = current directory	Created if needed, then made the process working directory during connection. Relative reference, map, model, and MDA paths are resolved from here. Changing the text after connection does not change directory.
Center wavelength (nm)	785.00; 0-2000	Desired spectrometer center wavelength. The bold label shows the hardware value. **Update wavelength** is enabled only after connection and applies the displayed wavelength to the spectrometer.
Grating	populated after connection	Lists the spectrometer gratings using 1-based indices. **Update grating** moves the turret and reports groove density and center wavelength. Wait for motion to finish before acquiring.

Connect hardware unloads old devices, loads the CFG, opens the napari-micromanager dock, creates the spectrum collector/DAQ, loads both models, and refreshes channels, wavelength, and gratings. **Reload transformer** reloads the Raman transformer and Vandermonde model without reconnecting. **Disconnect** unloads devices and clears all calibration, selection, writer, and model state; selection is prepared again after reconnecting.

Saved paths

The CFG, transformer, and Vandermonde fields point to existing input files. The **Output folder** becomes the base location for relative results. For example, `data/run` is resolved as `<Output folder>/data/run`.

Options at a glance

Option	Brief definition
Micro-Manager config	Hardware and device configuration file.
Transformer model	Bright-field-pixel to Raman-galvo coordinate model.
Vandermonde model	Image-pixel to stage-motion model for centering cells.
Output folder	Working directory applied when hardware connects.
Center wavelength	Spectrometer center wavelength setting.
Grating	Installed spectrometer grating selection.
Connect / Disconnect	Initializes or unloads the rig and session state.
Reload transformer	Reloads the coordinate models without reconnecting.

Collect Spectra Using Points Layer

Preparation: connect the collector and DAQ, load the transformer, and place a Points layer last. The first stored point is used for acquisition. Keeping one point in this layer makes the intended target clear.

Control	Default / allowed	Meaning and use
Exposure (ms)	1000; 1-1,000,000	Integration time used by the spectrum collector.
N (repeats, at least 2)	2; 2-1000	Number of identical copies of the transformed galvo coordinate used for acquisition. The minimum of two is enforced.
Save as	blank = display only	Optional filename. If the `.npy` suffix is provided, the widget adds it. A relative name is saved under the active working directory.

Collect spectra at last point restarts the galvo, transforms the point, collects N spectra, optionally saves the NumPy array, and opens a spectrum plot.

Saved output

When **Save as** contains a name, spectra are saved as `<output folder>/<name>.npy`. With a blank name, the spectra are displayed without creating a data file.

Options at a glance

Option	Brief definition
Exposure	Integration time for each Raman acquisition.
N (repeats)	Spectra acquired at the same transformed point.
Save as	Optional NumPy filename; blank disables saving.
Collect spectra	Transforms the point, acquires spectra, and displays them.

Laser Aiming Calibration and Recalibration

Control	Default / allowed	Meaning and use
N (repeats)	20; 1-1000	Spectra acquired per calibration target. More repeats increase acquisition time and can stabilize detection.
Exposure (ms)	1000; 1-1,000,000	Raman exposure used by the calibrator.
Max volts	1.80; 0.01-10.00	Galvanometer-voltage extent used by the calibration. Set it within the confirmed operating range of the rig.
Grid size	10; 2-100	Controls calibration sampling density. Larger grids require substantially more measurements.
Threshold	1.00; 0-100	Detection threshold used to accept/localize calibration responses. Its physical interpretation belongs to `cns-control.Calibrator`.
Recalibration	off	Reveals the controls for correcting the current calibration.
Model name	blank	Base filename for the corrected transformer. The save routine writes `<name>.json`, loads it immediately, and updates the Loading path.

Run calibration uses the active transformer and stores the resulting dataset in memory. It opens a log and calibration plot. Recalibration uses the most recent calibration dataset produced in the connected session.

For manual recalibration, check **Recalibration**, click **Open manual selector**, and then work through the displayed calibration frames: click a point; press Enter to advance; Backspace to return; R to reset; or N to mark a frame as NaN. Close or finish the selector, enter **Model name**, and click **Save recalibrated model**.

Saved output

The initial calibration remains in session memory for plotting and recalibration. **Save recalibrated model** writes `<output folder>/<Model name>.json` and immediately activates that model.

Options at a glance

Option	Brief definition
N (repeats)	Spectra used at each calibration target.
Exposure	Raman integration time during calibration.
Max volts	Galvanometer-voltage extent of the calibration.
Grid size	Sampling density of the calibration grid.
Threshold	Response criterion used by the calibrator.
Run calibration	Acquires and plots a new calibration dataset.
Recalibration	Reveals the manual correction controls.
Model name	Base filename for the corrected transformer JSON.
Open manual selector	Opens the last calibration dataset for correction.
Save recalibrated model	Saves, loads, and activates the corrected model.

Axial Background Scan

This section collects an autofocus/background Z series at one Raman point. Put a single-point Points layer last; the first point is transformed to galvo volts and repeated N times at every Z sample.

Control	Default / allowed	Meaning and use
Name	required; example `testing`	Human-readable prefix for the output.
Exposure (ms)	1000; 1-1,000,000	Exposure for each Raman spectrum.
N (spectra per z)	5; 1-1000	Replicate spectra at each axial position.
Search range (um)	10; 0.1-1000	Half-range around the starting Z; displayed coordinates span negative range to positive range. Verify objective clearance.
Search pts	20; 2-500	Number of axial samples across the full search interval.

Collect reference spectra performs the autofocus/background scan, moves the stage to the selected focus Z, plots all spectra, and writes `reference/<name>_<uuid>.zarr`. The dataset contains `spec[z,n,pixel]` plus Z, point X/Y, and exposure metadata. The stage is intentionally left at the found focus position.

Saved output

Each run is saved as `<output folder>/reference/<Name>_<uuid>.zarr`. The Zarr dataset stores the spectra, Z coordinates, selected point coordinates, and exposure value.

Options at a glance

Option	Brief definition
Name	Prefix identifying the saved reference dataset.
Exposure	Integration time for each reference spectrum.
N (spectra per Z)	Replicate spectra acquired at every Z position.
Search range	Positive and negative axial search extent.
Search points	Number of evenly spaced Z samples.
Collect reference spectra	Runs autofocus/background collection, plots, and saves.

Spatial Mapping

Create a Shapes layer, draw a rectangle, and make that layer last. The code uses only the first shape and reduces it to its min/max bounds; a rotated rectangle or nonrectangular polygon is therefore scanned as its axis-aligned bounding box.

Control	Default / allowed	Meaning and use
File name	required; example `scan_label`	Label inserted in the Zarr filename.
Raman exposure (ms)	1000; 1-1,000,000	Exposure at every Raman grid coordinate.
N (grid side)	20; 2-500	Produces an N by N grid, hence N squared Raman points per Z plane. Doubling N approximately quadruples point count.
Z offset (um)	4; -1000 to 1000	The base Raman Z is current Z minus this offset. The stage returns to its original Z after a successful scan.
Z-scan	off	When enabled, collects the full Raman grid at multiple Z planes.
Z range (+/- um)	5; 0.1-1000	Half-range around the base Raman Z; visible only for Z-scan.
Z steps	10; 2-500	Evenly spaced planes across the full Z range.
Extra channel	none; 1-1,000,000 ms	**+ Add channel** adds a Micro-Manager channel and exposure row; **x** removes it. Duplicate channels are ignored. BF is excluded here because BF is always acquired before and after the scan.

At start the widget snaps BF at 10 ms and one image for each extra channel. It then switches to RM, opens `Fluoshutter`, collects the Raman grid, closes the shutter, restores Z, and snaps ending BF. A Z-scan also records BF at every plane. Outputs are `grid_scan_data_<label>_<uuid>.zarr` or `grid_scan_z_<label>_<uuid>.zarr`, followed by a plot window.

Saved output

A single-plane map is saved as `<output folder>/grid_scan_data_<File name>_<uuid>.zarr`. A multi-plane map is saved as `<output folder>/grid_scan_z_<File name>_<uuid>.zarr`.

Options at a glance

Option	**Brief definition**
File name	Label inserted into the grid-scan filename.
Raman exposure	Integration time at every grid coordinate.
N (grid side)	X and Y sample count; total is N squared per plane.
Z offset	Raman-plane displacement from the current Z.
Z-scan	Enables multi-plane Raman acquisition.
Z range / Z steps	Axial half-range and number of sampled planes.
Add/remove channel	Manages fluorescence-channel and exposure pairs.
Run grid scan	Acquires BF, fluorescence, Raman grid data, and ending BF.

Generate Stage Grid

This section creates stage positions around the *current* stage XY and stores the prepared variables for **Run Raman MDA**. Every position carries the same fixed image point.

Control	Default / allowed	Meaning and use
Autofocus object	None, laser, software, quartz, glass, cell	Autofocus mode attached to the generated positions. The choice also controls which MDA autofocus fields are visible.
FOV x / FOV y (px)	740 / 540; 0-100000	Fixed image coordinate used at every field of view. Note the UI labels X then Y; internal selection receives both separately.
X range / Y range (+/- um)	100 / 100	Half-width of the stage grid about the current XY. Total spans are twice these values.
X step / Y step (um)	50 / 50; minimum 0.01	Stage-position spacing. Confirm that all generated positions remain within stage and sample limits.
Repeats (points per position)	2; 2-1000	Identical points at each stage position; minimum two is required by the DAQ.
Skip BF pre-scan	on	Uses blank placeholder images to establish dimensions. Turn off when real per-position BF images are needed for setup.

Generate grid stops live mode, prepares the napari-micromanager MDA sequence, builds the stage grid, and prepares `sources`, `autofocus_p`, and `new_seq`. After the status reports ``stage grid ready," proceed to **Run Raman MDA**.

Saved output

This section prepares variables in memory. The stage positions and Raman results are written later by **Run Raman MDA** to its **Writer output dir**.

Options at a glance

Option	**Brief definition**
Autofocus object	Focus strategy attached to generated positions.
FOV X / FOV Y	Fixed pixel target repeated at every position.
X/Y range	Grid half-widths around the current stage XY.
X/Y step	Stage spacing between neighboring positions.
Repeats	Identical aiming points carried by every position.
Skip BF pre-scan	Uses blank placeholders instead of BF setup images.
Generate grid	Creates positions and selection results for Raman MDA.

Automated Cell Selection

Mask controls

Control	Default / allowed	Meaning and use
Center Y / Center X	540 / 740; 0-100000	Center of the circular permitted region in image pixels, in Y,X order.
Radius	100; 1-100000	Radius of the permitted circular region. The same values are passed later to the Raman MDA engine.
Add mask	button	Adds a red masked overlay, with a green 10-pixel center marker, to help inspect the permitted region before selection.
Click to center	toggle button	Arms a one-shot viewer click that moves the stage so the clicked feature reaches Center Y/X. Requires a loaded Vandermonde model and connected hardware; use cautiously because it commands stage motion.

Selection controls

Control	Default / allowed	Meaning and use
Autofocus object	laser; choices laser, software, quartz, glass, cell, None	Autofocus strategy saved with the selection. None disables autofocus in the later MDA.
N per FOV	6; 1-1000	Requested cell count. Automated selection passes `N+1` to its helper. In manual batch mode it is the exact click count per FOV.
Center cell	off	Splits detections into one new stage position per cell and shifts each cell to the requested center. Requires the Loading Vandermonde model.
Aiming pattern	Square or Circle	Shape of the Raman subpoint pattern placed around each selected cell.
Pattern size (px)	30; 0-10000	Spatial extent of the point pattern.
N_x (subpoints)	1; 1-100	Pattern sampling parameter. Batch MDA requires the resulting pattern multiplier to be at least two.
Background distance (px)	80; 0-1,000,000	Distance threshold used by automated selection.
Integrated batch collection	False	In batch mode positions/click counts and Raman point handling differ; it must agree with dataset generation.
Cellpose model	installed models; defaults to cyto2 if available	Segmentation model used to identify candidate cells.

Run automated selection prepares the external MDA widget, performs Cellpose-based selection, creates source layers and a new sequence, and stores them for Raman MDA.

Manual selection creates empty point-source layers. In batch mode, click exactly **N per FOV** cells per FOV; in non-batch mode, click freely. The button immediately stores references to those layers, so finish clicking before starting the Raman MDA. For non-batch manual results, **Center clicked cells** uses the Vandermonde model to turn each clicked cell into a centered stage position, replacing the selection results.

Saved output

Automated and manual selection prepare `sources`, `autofocus_p`, and `new_seq` in memory. **Add mask** adds a napari layer in the current viewer. The subsequent Raman MDA writes the acquired data to **Writer output dir**.

Options at a glance

Option	Brief definition
Center Y / Center X	Pixel center of the permitted mask and centering target.
Radius	Radius of the circular permitted region.
Add mask	Displays the configured circular selection region.
Click to center	Moves a clicked feature to the configured center.
Autofocus object	Focus strategy saved with the selected positions.
N per FOV	Requested detections or manual batch clicks per field.
Center cell	Produces one centered stage position per detected cell.
Aiming pattern / size / N_x	Shape, extent, and sampling of Raman subpoints.
Background distance	Distance threshold for automated selection.
Integrated batch collection	Selects batch or individual point handling.
Cellpose model	Segmentation model used to detect cells.
Run automated selection	Segments cells and prepares MDA selection results.
Manual selection	Creates layers for hand-clicked cell selection.
Center clicked cells	Converts non-batch clicks into centered positions.

Run Raman MDA

Preparation: connect with both collector and transformer, then run automated selection, manual selection, centered-manual selection, or stage-grid generation. These workflows prepare the variables used by the MDA.

Control	Default / allowed	Meaning and use
Writer output dir	`data/run`	Directory where Raman TIFF and NumPy outputs are written; blank uses `data/run`.
Autofocus positions	blank = selection	Optional comma-separated zero-based stage position indices, e.g. `0,2,5`. `None` behaves like blank. Out-of-range indices are rejected.
Imaging positions	blank = autofocus positions	Optional comma-separated zero-based indices at which imaging occurs. The code initializes this from the original selection autofocus list before applying overrides, so set this explicitly when using a substantially different autofocus override.
Raman glass offset (um)	5; -1000 to 1000	Axial offset used for the Raman measurement.
Autofocus search range	6; 0.1-1000	Coarse autofocus range; hidden if autofocus is None.
Autofocus search pts	8; 2-500	Coarse autofocus sample count.
Laser fine search range (+/- um)	1.5; 0.1-1000	Fine range shown only for laser autofocus.
Laser fine search pts	8; 2-500	Fine laser-autofocus sample count.
Segment and track	off	Re-segments images and updates aiming during a time series.
Segment channel	BF initially	Micro-Manager channel used for segmentation.
Image rescale factor	2.0; 0.1-100	Image scale used during segmentation.
Cellpose model	installed models; cyto2 preferred	Model for time-series re-segmentation.
Crop image around mask	True	Whether segmentation is cropped to the circular mask region.
Tracking config (.json)	`particle_config.json`	Particle-tracking configuration.
Exposure per cell (ms)	1000; 1-1,000,000	Total exposure supplied while building the Raman sequence. In non-batch mode it is multiplied by the aiming-pattern multiplier.
Loops (time points)	100; 1-1,000,000	Number of temporal repetitions.
Interval (s)	600; 0-1,000,000	Requested interval between time points.
Refocus & segment every	1	Cadence in time points; 1 means every time point.
Z relative (comma-separated um)	`0, 4`	Relative Z planes used to replace the sequence Z plan. Values must parse as floats.
Raman z indices	`0`	Zero-based indices into the Z list where Raman is requested. Ensure every index exists; e.g. with two Z values, valid indices are 0 and 1.
Extra fluorescence channel	none; default row exposure 10 ms	**+ Add channel** adds any Micro-Manager channel (including BF); **x** removes it. Duplicate channel names are ignored. Added channels inherit the first sequence channel as a template.

The acquisition uses axis order `t,p,c,z`, assigns the selection's aiming-source layers, replaces the time and Z plans, adds unique fluorescence channels, and writes Raman metadata containing the requested Z indices. **Stop** requests MDA cancellation and stops sequence acquisition; the current hardware event may finish before exit.

Generate dataset

Generate dataset asks for a completed MDA run directory. It loads the experiment, using the current **Integrated batch collection** value, and writes sibling outputs `data/dataset/ds_<run>.zarr` and `df_<run>.pkl`. It also opens the dataset viewer. Set the batch dropdown to match how that run was acquired before generating.

Pixel-to-stage calibration

Check **Pixel-to-stage calibration** to reveal these controls:

Control	Default / allowed	Meaning and use
Dataset (.zarr)	required	Generated dataset containing a JSON `useq_sequence` attribute and one image per stage position.
Vandermonde degree	1; 1-5	Polynomial degree. Degree d needs at least (d+1)(d+2)/2 valid picked points. Prefer the lowest degree with adequate residuals.
Pick points	button	Opens images position by position. Click the same recognizable feature in each frame; skipped/NaN frames are excluded.
Model file	`vandermonde_model.json`	Suggested save name. A save dialog still asks for the final location.
Fit & save model	button	Centers image and stage coordinates, reports degree 1-3 RMSE where possible, fits the selected degree, and saves JSON. The saved path is also copied into Loading's Vandermonde field for immediate center-cell use.

Saved output

The Raman acquisition is saved under `<output folder>/<Writer output dir>/`. The writer creates the TIFF and NumPy files for the run. **Generate dataset** saves `<run parent>/dataset/ds_<run>.zarr` and `<run parent>/dataset/df_<run>.pkl`. Pixel-to-stage calibration saves the chosen **Model file** as JSON at the location selected in the save dialog.

Options at a glance

Option	**Brief definition**
Writer output directory	Destination used by the Raman TIFF/NumPy writer.
Autofocus positions	Zero-based position indices that receive autofocus.
Imaging positions	Zero-based position indices that receive imaging.
Raman glass offset	Axial offset between glass focus and Raman measurement.
Autofocus range / points	Extent and sampling count of the coarse focus search.
Laser fine range / points	Extent and sampling count of laser fine focus.
Segment and track	Re-detects cells and updates aiming during the time series.
Segment channel	Image channel used for segmentation.
Image rescale factor	Resampling factor supplied to segmentation.
Cellpose model	Model used for repeated cell segmentation.
Crop around mask	Restricts segmentation to the configured mask region.
Tracking config	JSON settings used for cell/particle tracking.
Exposure per cell	Raman exposure budget assigned to each selected cell.
Loops / interval	Number of time points and requested spacing between them.
Refocus & segment every	Time-point cadence for focus and tracking updates.

Option	Brief definition
Z relative	Comma-separated relative Z planes in micrometres.
Raman Z indices	Indices of relative Z planes receiving Raman acquisition.
Add/remove fluorescence channel	Manages channel/exposure pairs in the sequence.
Run Raman MDA	Builds and launches the final `t,p,c,z` acquisition.
Stop	Requests MDA cancellation and stops sequence acquisition.
Generate dataset	Converts a completed run into Zarr and pickled-table outputs.
Pixel-to-stage calibration	Reveals the Vandermonde calibration workflow.
Dataset	Zarr images and stage coordinates used for calibration.
Vandermonde degree	Polynomial degree used for the pixel-to-stage fit.
Pick points	Opens the common-feature picker for each stage position.
Model file	Suggested JSON filename for the fitted model.
Fit & save model	Fits, reports residuals, saves, and activates the model.

Session Checklist and Output Summary

1. Verify the CFG, Raman transformer, optional Vandermonde model, and output folder.
2. Connect; confirm the status, wavelength, grating, and available channels.
3. Verify the intended layer is last before point or shape operations.
4. For MDA work, configure the napari-micromanager MDA widget and run exactly one selection-producing workflow. Finish manual clicks before continuing.
5. Check all physical motion ranges, point counts, Z planes, loops, and exposures.
6. Run a small pilot, inspect the log and plots, then scale to the full acquisition.

Workflow	Output
Point spectrum	Optional `<name>.npy`; plot window.
Calibration	In-memory dataset and plot; recalibration saves `.json`.
Axial background scan	`reference/<name>_<uuid>.zarr`.
Spatial mapping	`grid_scan_data_*.zarr` or `grid_scan_z_*.zarr`.
Selections / stage grid	In-memory source layers, autofocus indices, and MDA sequence.
Raman MDA	TIFF/NumPy writer directory under **Writer output dir**.
Generate dataset	`ds_<run>.zarr` and `df_<run>.pkl`.
Pixel-to-stage fit	Vandermonde JSON model.

Source: https://github.com/JasonYu1/napari-raman-widget.